

Solucion-Ejercicio-1.pdf



Anónimo



Ingenieria Estructural



1º Máster en Ingeniería Industrial



Escuela Politécnica Superior. Campus de Leganés
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antes



**Descarga sin publi
con 1 coin**



Después

WUOLAH

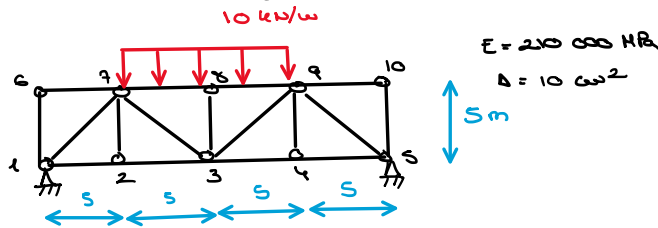


Importante

Puedo eliminar la publi de este documento con 1 coin

¿Cómo consigo coins? → Plan Turbo: barato
→ Planes pro: más coins

Solución ejercicio 1

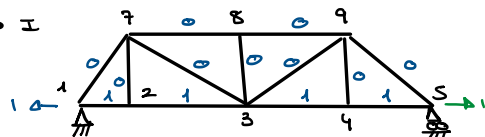


$$CH = 17 + 4 - 2 \cdot 10 = 1, \text{ Hipereestática.}$$

1. Determinar los esfuerzos de todas las barras.

- Barras 1-6 + 6-7 + 5-10 + 9-10 las podemos eliminar, ya que no están trabajadas.

Estado I



$$\sum F_H = 0; H_1 + 1 = 0; H_1 = -1$$

$$\sum F_V = 0; V_1 + V_S = 0; V_1 = 0$$

$$\sum H_1 = 0; V_S \cdot 20 = 0; V_S = 0$$

Nudo 1:



$$N_{17} \frac{\sqrt{2}}{2} = 0; N_{17} = 0$$

$$N_{12} = 1$$

Nudo 2:



$$N_{29} = 0$$

$$N_{23} = 1$$

Nudo 7:

$$\rightarrow N_{78}$$

$$\downarrow N_{73}$$

$$N_{73} = 0$$

$$N_{78} = 0$$

Nudo 8:

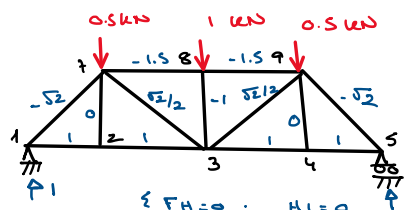
$$\rightarrow N_{89} = 0$$

$$\downarrow N_{83}$$

$$N_{83} = 0$$

⊕

Estado II



$$\sum F_H = 0; H_1 = 0$$

$$\sum F_V = 0; V_1 + V_S = 2; V_1 = 1$$

$$\sum H_1 = 0; -0.5 \cdot 5 - 1 \cdot 10 - 0.5 \cdot 15 + V_S \cdot 20 = 0$$

$$V_S = 1$$

Nudo 1:



$$N_{17} \frac{\sqrt{2}}{2} + 1 = 0$$

$$N_{17} = -\sqrt{2}$$

$$N_{12} + N_{17} \frac{\sqrt{2}}{2} = 0$$

$$N_{12} = 1$$

Nudo 7:



$$-N_{73} \frac{\sqrt{2}}{2} + \sqrt{2} \cdot \frac{\sqrt{2}}{2} - 0.5 = 0$$

$$N_{73} = \sqrt{2}/2$$

$$N_{73} \frac{\sqrt{2}}{2} + N_{78} + 1 = 0$$

$$N_{78} = -1.5$$

Nudo 8:



$$N_{83} = -1$$

		$H_S \cdot N_i^I + S_0 N_i^II$					
Barra	Li	N_i^I	N_i^II	$(N_i^I)^2 Li$	$(N_i^II \cdot N_i^I) Li$	N_i^0	
9-5 1-7	$5\sqrt{2}$	0	$-\sqrt{2}$	0	0	$-50\sqrt{2}$	
4-5 1-2	S	1	1	$S \times 2$	$S \times 2$	0	
4-9 2-7	S	0	0	0 <i>barra</i>	0	0	
3-9 7-3	$5\sqrt{2}$	0	$\sqrt{2}/2$	0	0	$50\sqrt{2}/2$	
4-3 2-3	S	1	1	$S \times 2$	$S \times 2$	0	
8-9 7-8	S	0	-1.5	0 <i>barra</i>	0	-7.5	
8-3	S	0	-1	0	0	-50	
				20	20		

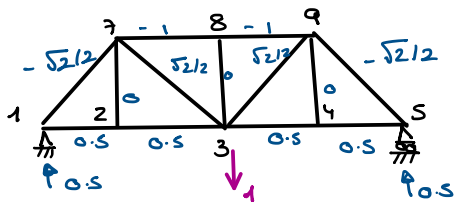
$$V_S = \frac{\sum (S_0 \cdot N_i^II + H_S \cdot N_i^I) \cdot N_i^I \cdot Li}{EA} = 0$$

$$S_0 \cdot 20 + H_S \cdot 20 = 0$$

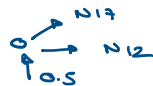
$$H_S = -50 \text{ kN} //$$

2. Movimiento vertical del nodo 3.

Estado III



Nodo 1:



$$N_{17} \frac{\sqrt{2}}{2} + 0.5 = 0$$

$$N_{17} = -\frac{\sqrt{2}}{2}$$

$$N_{12} + N_{17} \frac{\sqrt{2}}{2} = 0$$

$$N_{12} = 0.5$$

Nodo 2:



$$N_{27} = 0$$

$$N_{23} = 0.5$$

Nodo 7:



$$-N_{73} \frac{\sqrt{2}}{2} + \frac{1}{\sqrt{2}} \frac{\sqrt{2}}{2} = 0$$

$$N_{73} = \frac{\sqrt{2}}{2}$$

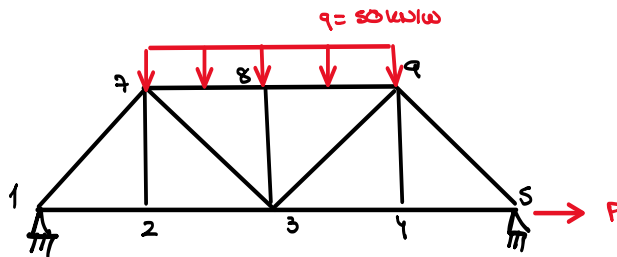
$$N_{78} + N_{73} \frac{\sqrt{2}}{2} + \frac{1}{\sqrt{2}} \frac{\sqrt{2}}{2} = 0$$

$$N_{78} = -1$$

Barros	L_i	N_i^0	N_i^I	$N_i^0 \cdot N_i^I \cdot L_i$
9-5 1-7	$5\sqrt{2}$	$-50\sqrt{2}$	$-\sqrt{2}/2$	$(250\sqrt{2}) \times 2$
4-5 1-2	5	0	0.5	0
4-9 2-7	5	0	0	0
3-9 7-3	$5\sqrt{2}$	$50\sqrt{2}/2$	$\sqrt{2}/2$	$(125\sqrt{2}) \times 2$
4-3 2-3	5	0	0.5	0
9-9 7-8	5	-75	-1	$(375) \times 2$
8-3	5	-50	0	0

$$V_3 = \frac{2 \cdot (905.33) \cdot 10^3 (N \cdot m)}{210.000 \cdot 10^6 \cdot 0.001 (m^2) (Pa)} = 8.622 \cdot 10^{-3} //$$

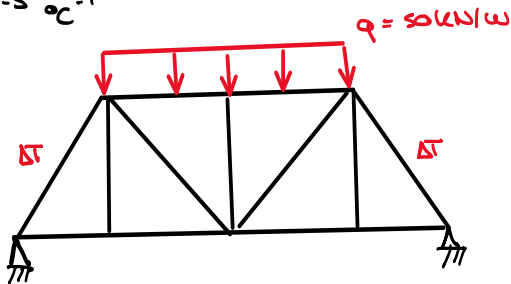
3. Calcular fuerza F en nodo 3 para que $U_3 = 0$.



Caso la estructura es hiperestática y la carga está aplicada en S, la carga F va a ser absorbida por el apoyo y el nodo 3 va a seguir desplazándose verticalmente. La única forma de que $U_3 = 0$ para este caso, es que $q = 0$.

4. Valor del incremento ΔT en barras 1-7 y 9-5 para $U_3 = 0$.

$$\alpha = 1 \cdot 10^{-5} \text{ } ^\circ\text{C}^{-1}$$



Barros	L_i	δT	N_i^I	$\delta T N_i^I$
1-7	$5\sqrt{2}$	$\Delta T \alpha$	$-\sqrt{2}/2$	$-\Delta T \alpha 5$
9-5	$5\sqrt{2}$	$\Delta T \alpha$	$-\sqrt{2}/2$	$-\Delta T \alpha 5$

$$V_3 = 0.00862 - \Delta T \alpha \cdot 10 = 0$$

$$\Delta T = 86.2 \text{ } ^\circ\text{C} //$$